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(Deemed to be University)

Assignment

Branch	CSE	Program	B.Tech.
Subject Name	Data Mining and Predictive Analytics	Semester	VII
Subject Code	CSE4859	Year	2026

Theory Assignment 1

Date of Submission: On or before 05/10/2026

Learning Level (LL)	L1: Remembering	L3: Applying	L5: Evaluating
	L2: Understanding	L4: Analyzing	L6: Creating

Q. No.	Question	COs	LL
1	a) Define Data Mining. Describe the steps involved in knowledge discovery process from data. b) Compare and contrast Predictive Analytics with Data Mining.	CO1	L1, L2
2	Discuss the need for human direction of data mining. Describe the possible consequences of relying on completely automatic data analysis tools.	CO1	L2, L4
3	For each of the following, identify the relevant data mining task(s): a) The Boston Celtics would like to approximate how many points their next opponent will score against them. b) A NORAD defense computer must decide immediately whether a blip on the radar is a flock of geese or an incoming nuclear missile. c) A political strategist is seeking the best groups to canvass for donations in a particular county. d) A Homeland Security official would like to determine whether a certain sequence of financial and residence moves implies a tendency to terrorist acts. e) Wall Street analyst has been asked to find out the expected change in stock price for a set of companies with similar price/earnings ratios	CO1	L3
4	For each of the following meetings, explain which phase in the CRISP-DM process is represented: a) Managers want to know by next week whether deployment will take place. Therefore, analysts meet to discuss how useful and accurate their model is. b) The data mining project manager meets with the data warehousing manager to discuss how the data will be collected. c) The data mining consultant meets with the Vice President for Marketing, who says that he would like to move forward with customer relationship management. d) The data mining project manager meets with the production line supervisor, to discuss implementation of changes and improvements.	CO1	L3

	e) The analysts meet to discuss whether the neural network or decision tree models should be applied.																				
5	a) Are noise and outlier same? Explain with an example. b) Why do we need to treat outliers carefully?	CO2	L2																		
6	a) Describe the possible negative effects of proceeding directly to mine data that has not been preprocessed. b) Explain why it is not recommended, as a strategist for dealing with missing data, to simply omit the records or fields with missing values from the analysis. c) Explain why data analysts need to normalize their numeric variables.	CO2	L1, L2																		
7	Consider following stock price data: <table border="1"><tr><td>Stock Price</td><td>10</td><td>7</td><td>20</td><td>12</td><td>75</td><td>15</td><td>9</td><td>18</td><td>4</td><td>12</td><td>8</td><td>14</td></tr></table> a) Compute mean, median, mode of the given data. Comment on modality of the data. b) Compute the midrange stock price. c) Compute min-max normalized stock price for the stock worth \$20. d) Compute Z-score normalized stock price for the stock worth \$20. e) Compute decimal standardized stock price for the stock worth \$20 f) Compute the skewness of the given stock price data. g) Find outlier in the given data using IQR method. h) Find outlier in the given data using Z-score method.	Stock Price	10	7	20	12	75	15	9	18	4	12	8	14	CO2	L3					
Stock Price	10	7	20	12	75	15	9	18	4	12	8	14									
8	Find five number summary for the following data, and draw a box-and-whisker plot: {2,6,7,8,8,11,12,13,14,15,22,23}.	CO2	L3																		
9	Consider the following attribute and their respective value. Identify the attribute type based on its value. <table border="1"><tr><td></td><td>Attribute Name</td><td>Value</td></tr><tr><td>a)</td><td>Student Name</td><td>Kabir</td></tr><tr><td>b)</td><td>Registration ID</td><td>ITERCSE23001</td></tr><tr><td>c)</td><td>Date of Birth</td><td>01/01/2005</td></tr><tr><td>d)</td><td>CGPA</td><td>9.4</td></tr><tr><td>e)</td><td>Class Rank</td><td>5</td></tr></table>		Attribute Name	Value	a)	Student Name	Kabir	b)	Registration ID	ITERCSE23001	c)	Date of Birth	01/01/2005	d)	CGPA	9.4	e)	Class Rank	5	CO2	L3
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10	Consider the following data: 11, 13, 13, 15, 15, 16, 19, 20, 20, 20, 21, 21, 22, 23, 24, 30, 40, 45, 45, 45, 71, 71, 73, 75 a) Partition the data using equal frequency (Equi-depth) partitioning with bin depth of 6. b) Partition the data using equal width partitioning. c) Use smoothing by bin mean, median, and boundaries for a) and b)		L3																		